

The Skew Risk Premium

Implied versus Realized Asymmetry, and Where the Moment Ladder Breaks

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Abstract

The variance risk premium says that implied volatility exceeds realized volatility on average, and that the gap is tradeable. This note carries the same logic one moment up. Implied skew (the asymmetry the option surface prices) is more extreme than the asymmetry returns actually realize, and the difference is a second premium, independently tradeable and driven by its own forces. We fix the two distinct meanings of “skew” that this subject constantly conflates, define the premium precisely, and derive the model-free link between the option surface and risk-neutral skewness through Bakshi–Kapadia–Madan. We then ask what sustains the premium, how noisily its realized leg can be measured, how it is traded, and (the theme the note builds toward) why the same programme, pushed to the fourth moment and beyond, stops paying for itself. The moment ladder is not infinite: each rung is noisier to measure and more fragile to extract than the last, and the note is as much about where that ladder breaks as about the rung that works.

Contents

I	Trading the skew: decomposing the skew risk premium	3
1	Two meanings of “skew”	3
1.1	Implied skew: a slope on the surface	3
1.2	Realized skew: a moment of returns	5
1.3	Why the surface slope encodes distributional asymmetry	5
2	Defining the skew risk premium	6
2.1	The premium	6
2.2	Sign and economic reading	7
3	Model-free extraction: Bakshi–Kapadia–Madan (BKM)	7
3.1	From variance replication to arbitrary moments	7
3.2	The kernels, and what their shape already tells us	7
3.3	The moment formulas	8
4	The realized leg is noisy	9
4.1	The standard error of sample skewness	10
4.2	Robust alternatives	11
4.3	Window length	12

5	What sustains the premium	12
5.1	Flow: structural demand for the put wing	13
5.2	Jumps, and why realized skew under-counts them	13
5.3	Correlation: the index-only term	13
6	Trading the premium	13
6.1	The canonical vehicle: the risk reversal	13
6.2	The Greek profile, done carefully	14
6.3	What is really being traded	14
6.4	Other vehicles, and an honest limit	14
6.5	Four situations of trading skew	15
7	Questions the framework should answer	16
II	Remarks about trading other moments	17
8	Why variance, not volatility	17
8.1	Variance is additive; volatility is not	17
8.2	Variance replicates cleanly; volatility does not	17
8.3	The square root carries a Jensen wedge	18
8.4	The VIX methodology made exactly this switch	18
9	Where the moment ladder breaks	18
9.1	Measurement noise doubles with each rung	19
9.2	Truncation bias worsens, and then inverts the sign	19
9.3	The premium stops being distinct	20
10	Synthesis	21

Part I

Trading the skew: decomposing the skew risk premium

1 Two meanings of “skew”

The word “skew” names two different objects in options, and almost every confusion on this subject is a collision between them. One is a property of the option surface; the other is a property of the return distribution. They are related (that relation is the whole point) but they are not the same quantity, and keeping them apart is the first discipline.

1.1 Implied skew: a slope on the surface

The *implied skew* is a feature of the volatility surface: the tilt of the smile, that is, of implied volatility plotted against strike at a fixed maturity. Like the tilt of any curve, it can be measured in two ways (at a single point, or between two points) and both are in use.

At a single point, the skew is the slope of the tangent to the smile at the money. Measured in log-moneyness $k = \log(K/F)$,

$$\psi(T) = \left. \frac{\partial \sigma_{IV}}{\partial k} \right|_{k=0}. \quad (1)$$

Log-moneyness rather than raw strike is deliberate: a slope per dollar of strike depends on the price level, so a ten-dollar stock and a five-hundred-dollar stock with identically shaped smiles would report slopes fifty times apart. In log-moneyness they report the same number. For equities $\psi < 0$: implied volatility falls as the strike rises.

Between two points, the skew is the slope of the secant joining two strikes on either side of the money. The market standard takes the 25-delta put and the 25-delta call and divides their vol spread by the ATM level, giving the normalized risk reversal

$$\mathcal{S}(T) = \frac{\sigma_{IV}(25\Delta \text{ put}, T) - \sigma_{IV}(25\Delta \text{ call}, T)}{\sigma_{IV}(\text{ATM}, T)}. \quad (2)$$

The tangent is the cleaner mathematical object; the secant is what a desk quotes, because it is read directly from two traded options rather than differentiated out of a fitted curve.

We fix the sign convention once and hold it for the whole note: with puts minus calls in the numerator, $\mathcal{S} > 0$ is the ordinary equity regime (OTM puts richer in vol than OTM calls) and $\mathcal{S} < 0$ is an inverted, call-skewed regime. (The desk “risk reversal” is often quoted the other way, calls minus puts, and is then negative for equities; it is the same information with the sign flipped. To avoid the sign trap entirely, trades are described by which wing is sold, not by the sign of a number.) The ATM normalization is equally deliberate: it strips out the level of volatility, so that a rise in overall vol is not misread as a steepening of skew.

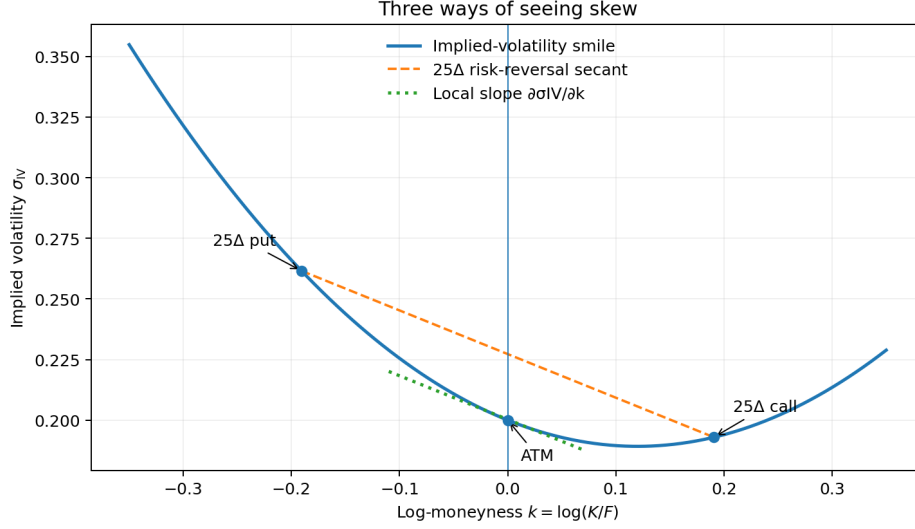


Figure 1: **Three ways of seeing skew.** The same implied-volatility surface can be described at different levels of resolution. The local slope $\psi = \partial\sigma_{IV}/\partial k$ measures the instantaneous change in implied volatility around the forward, while the 25Δ risk reversal measures the difference between two points on the smile. The two are related, but they are not identical statistics. The risk reversal therefore provides a practical desk-level measure of skew, whereas the local slope is a local property of the surface. More fundamentally, both are surface summaries of a risk-neutral distribution rather than direct measurements of realized asymmetry.

How the two measures connect. Where do the 25-delta strikes sit? To a first approximation, an option's delta is the probability that it finishes in the money, so a 25-delta call is struck where there is about a one-in-four chance of the underlying ending above it. For a normal distribution, one quarter of the probability lies more than 0.674 standard deviations above the mean. The 25-delta call therefore sits about 0.674 standard deviations above the forward, and by symmetry the 25-delta put about 0.674 below (a gap of 1.35 standard deviations between them). One standard deviation of the log return to expiry is $\sigma\sqrt{T}$, so the gap is $1.35\sigma\sqrt{T}$ in log-moneyness. Across that gap the smile falls at slope ψ , so the put vol exceeds the call vol by about $-\psi \times 1.35\sigma\sqrt{T}$. Dividing by the ATM vol, the σ cancels:

$$\mathcal{S}(T) \approx -1.35 \psi(T) \sqrt{T}.$$

Two consequences. The ATM normalization is what makes the risk reversal a clean function of the slope, since it exactly cancels the volatility in the strike gap. But $\sqrt{T}$ survives: for a fixed slope, the risk reversal grows with maturity, so comparing $\mathcal{S}$ across tenors does not compare slopes. The approximation is accurate to within a few percent at short tenors and loosens at high volatility and long maturity, where the delta-to-probability link is less exact.

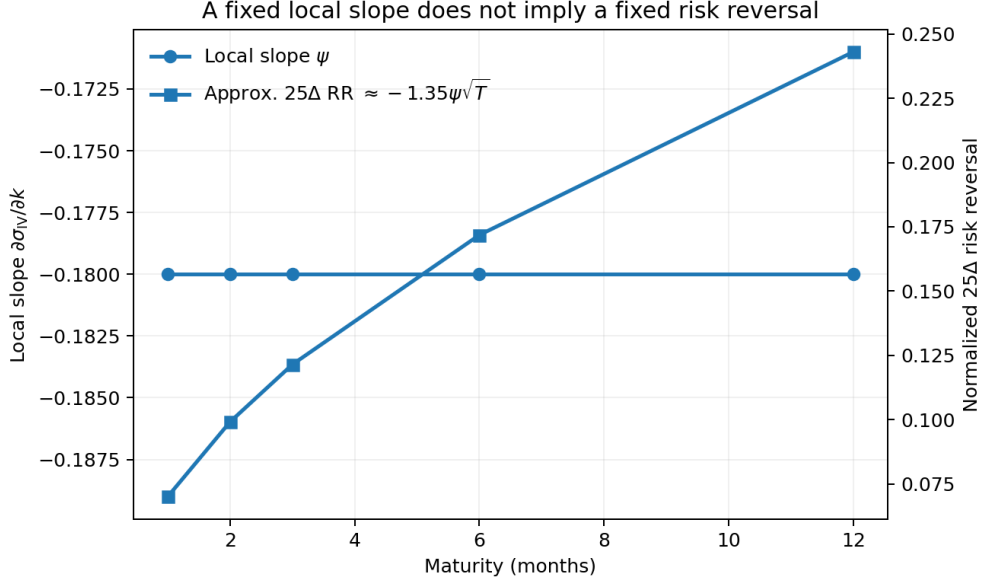


Figure 2: **A fixed local slope does not imply a fixed risk reversal.** For a locally linearized smile, the 25Δ risk reversal satisfies approximately $S(T) \approx -1.35 \psi(T) \sqrt{T}$. Consequently, even if the local slope ψ were unchanged, the magnitude of the risk reversal would increase with maturity. A comparison across tenors must therefore account for the different strike distance represented by a given delta and for the $\sqrt{T}$ scaling implicit in the approximation.

1.2 Realized skew: a moment of returns

The *realized skew* is a property of the return distribution: the third standardized moment of realized log returns over a window,

$$\text{Skew}_{[0,T]}^{\mathbb{P}} = \frac{\mathbb{E}^{\mathbb{P}}[(r - \mu)^3]}{(\mathbb{E}^{\mathbb{P}}[(r - \mu)^2])^{3/2}}, \quad (3)$$

with r the log return and μ its mean. It is dimensionless and it is a $\mathbb{P}$ -measure object (a statement about how returns actually came out, estimated from history).

1.3 Why the surface slope encodes distributional asymmetry

The two are tied because the surface slope is a low-dimensional summary of the risk-neutral density's asymmetry. By Breeden–Litzenberger, the risk-neutral density is the discounted second strike-derivative of the call price,

$$q(K | T) = e^{rT} \frac{\partial^2 C(K, T)}{\partial K^2}. \quad (4)$$

A steep put skew (higher implied vol at low strikes) is exactly the statement that q carries more mass in its left tail. The slope of σ_{IV} in K and the asymmetry of q cannot exist independently; one is a reparametrization of the other. This is why implied skew is a legitimate proxy for risk-neutral skewness: it is an observable, one-number handle on the distributional asymmetry that q encodes in full.

From the surface and return history to the skew risk premium

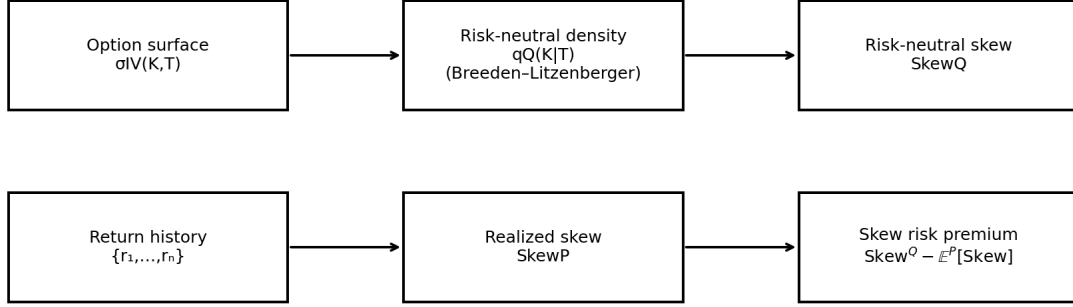


Figure 3: **From the option surface and return history to the skew risk premium.** The theoretical skew risk premium compares the risk-neutral expectation of standardized asymmetry with the corresponding physical expectation, $SRP = Skew^Q - \mathbb{E}^P[Skew]$. The option surface contains risk-neutral pricing information and can be mapped into distributional moments, while return history provides an estimate of physical, or realized, asymmetry. In empirical applications, a surface statistic such as a 25Δ risk reversal may be compared with realized skewness as a proxy for this theoretical premium. The two quantities should not be interpreted as mathematically identical.

Implied skew (a slope on the surface) and realized skew (a moment of returns) measure the same phenomenon (distributional asymmetry) from opposite sides. The surface slope summarizes the *risk-neutral* asymmetry; the third moment measures the *physical* asymmetry. The gap between them is the skew risk premium, and everything in this note is about that gap.

2 Defining the skew risk premium

2.1 The premium

The skew risk premium is the difference between the asymmetry the surface prices and the asymmetry returns are expected to realize:

$$SRP_{t,T} = Skew_t^Q [r_{[t,T]}] - \mathbb{E}_t^P [Skew_{[t,T]}], \quad (5)$$

the risk-neutral skewness (read from the option surface, §3) minus the expected realized skewness (forecast from history). It is the third-moment analogue of the variance risk premium, and it inherits the VRP's structure exactly: a $\mathbb{Q}$ -object minus a $\mathbb{P}$ -object, positive-priced because investors pay to be protected against the tail it describes.

In practice the clean moment form of (5) is often replaced by a surface proxy (the current normalized risk reversal $\mathcal{S}(T)$ against a rolling realized-skew estimate, rescaled to comparable units. The moment form is the theoretically clean object (both legs are dimensionless third moments); the surface form is what a desk actually computes ($\mathcal{S}$ is directly observable, sample skew is a one-line calculation). They agree in spirit and we use whichever is clearer in context.

2.2 Sign and economic reading

For equity indices both legs are negative, and the risk-neutral leg is the more negative. Representative one-month figures are a risk-neutral skewness around -1.5 to -2.5 for the index against a realized skewness around -0.3 to -0.8 in calm regimes, so the premium $\text{SRP} = \text{Skew}^{\mathbb{Q}} - \text{Skew}^{\mathbb{P}}$ sits around -1 to -2 . The premium is structurally negative: the surface prices a more extreme left-tail asymmetry than returns typically deliver. Economically this is crash insurance sold rich — investors overpay, in distributional terms, for downside protection relative to the asymmetry that actually materializes, and the seller of that protection earns the gap.

The variance premium is the *level* premium: implied vol exceeds realized vol. The skew premium is the *asymmetry* premium: implied skew is more negative than realized skew. Both are risk-averse investors paying up for protection against adverse outcomes; both are tradeable, but through different vehicles and with different tails (as later sections make precise, the skew trade is a vanna trade where the variance trade is a gamma trade).

3 Model-free extraction: Bakshi–Kapadia–Madan (BKM)

The premium's risk-neutral leg needs a number for $\text{Skew}^{\mathbb{Q}}$. Bakshi, Kapadia and Madan (2003) supply it directly from option prices, with no model of the dynamics assumed. The construction is the one the variance note already built, run at higher order: the fair variance derived there is the second moment, and the third and fourth follow from the same static replication with different weights.

3.1 From variance replication to arbitrary moments

The variance note showed that any twice-differentiable payoff $H(S_T)$ is replicated by cash, a forward position, and a strip of out-of-the-money options weighted by its own second derivative $H''(K)$. That result is used here rather than re-derived: its content is that the price of H is fixed by the option surface through

$$\mathbb{E}^{\mathbb{Q}}[H(S_T)] = (\text{cash and forward}) + \int_0^{S_0} H''(K) P(K) dK + \int_{S_0}^{\infty} H''(K) C(K) dK, \quad (6)$$

with each option weighted by H'' at its strike.

The variance swap is the case $H(S_T) = -2 \log(S_T/S_0)$, whose second derivative $H''(K) = 2/K^2$ is the flat $1/K^2$ weighting of that note. To reach the higher moments, take instead the raw powers of the log return, $H(S_T) = r^n$ with $r = \log(S_T/S_0)$, for $n = 2, 3, 4$. Nothing about the machinery changes; only the weight does, because only H'' does.

3.2 The kernels, and what their shape already tells us

Differentiating r^n twice and evaluating at the strike gives the weight each option carries in the n -th contract:

$$r^2 : \frac{2(1 - \log \frac{K}{S_0})}{K^2}, \quad r^3 : \frac{6 \log \frac{K}{S_0} - 3 \log^2 \frac{K}{S_0}}{K^2}, \quad r^4 : \frac{12 \log^2 \frac{K}{S_0} - 4 \log^3 \frac{K}{S_0}}{K^2}. \quad (7)$$

Each is the flat $1/K^2$ of the variance swap multiplied by a polynomial in $\log(K/S_0)$ of degree $n - 1$, and that polynomial is the whole story of what distinguishes the moments. The variance kernel

($n = 2$) is even in $\log(K/S_0)$: it weights the two wings symmetrically, which is why variance does not care about the direction of a move. The skewness kernel ($n = 3$) is odd: puts and calls enter with opposite sign, which is exactly what lets it measure asymmetry, and is why the put integral is subtracted in the contract below. The kurtosis kernel ($n = 4$) carries $\log^2$ and $\log^3$, so it grows faster than either as K leaves the money and loads the far wings hardest of all.

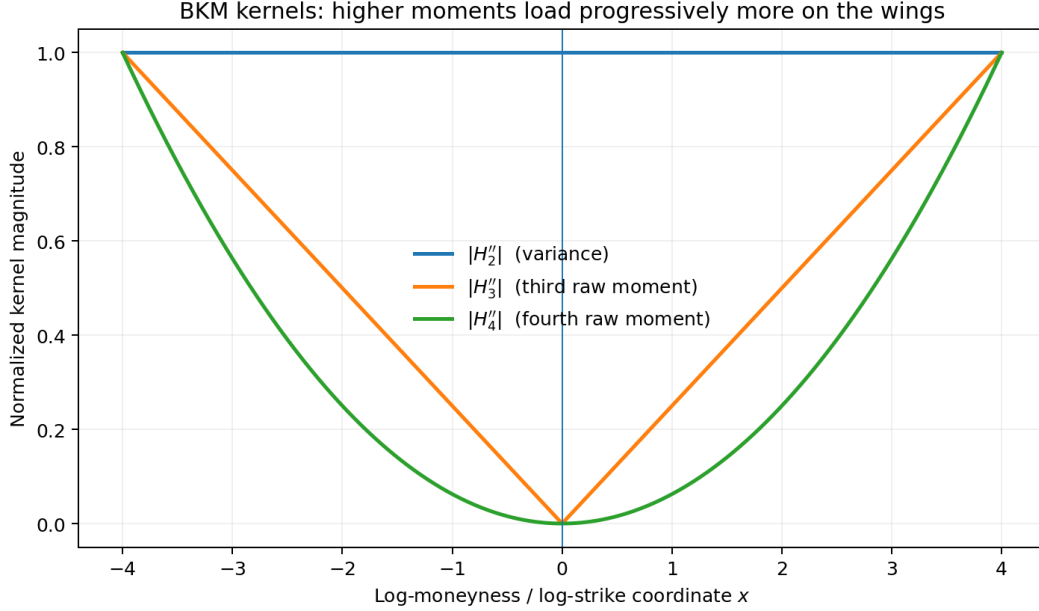


Figure 4: **Higher moments place progressively greater weight on the wings.** The figure shows the qualitative shape of the second-derivative kernels appearing in the Bakshi–Kapadia–Madan representation of terminal log-return moments. The variance kernel is comparatively uniform, whereas the kernels associated with the third and fourth moments grow in magnitude away from the center. The implication is important: extracting higher moments from option prices requires increasingly reliable information from deep out-of-the-money options. The figure is schematic and illustrates relative wing sensitivity rather than the exact replication weights.

Remark 3.1 (The shape is the source of the later trouble). Read the degrees off (7): the weight far from the money grows like $\log^{n-1}(K/S_0)$, so each higher moment leans more heavily on the wings than the last. This is not an aside. It is the reason the truncation of a real, finite strike range biases skewness toward zero and kurtosis further still (the moments that need the wings most are the ones a truncated surface reports worst. The kernel shape here is the mechanism of §4.

3.3 The moment formulas

Pricing the three payoffs (7) against the surface defines the quadratic, cubic and quartic contracts.

Theorem 3.2 (Bakshi–Kapadia–Madan, 2003). For maturity T , with the OTM option prices

$C(K), P(K),$

$$V = \int_{S_0}^{\infty} \frac{2(1 - \log(K/S_0))}{K^2} C(K) dK + \int_0^{S_0} \frac{2(1 + \log(S_0/K))}{K^2} P(K) dK, \quad (8)$$

$$W = \int_{S_0}^{\infty} \frac{6 \log(K/S_0) - 3 \log^2(K/S_0)}{K^2} C(K) dK - \int_0^{S_0} \frac{6 \log(S_0/K) + 3 \log^2(S_0/K)}{K^2} P(K) dK, \quad (9)$$

$$X = \int_{S_0}^{\infty} \frac{12 \log^2(K/S_0) - 4 \log^3(K/S_0)}{K^2} C(K) dK + \int_0^{S_0} \frac{12 \log^2(S_0/K) + 4 \log^3(S_0/K)}{K^2} P(K) dK. \quad (10)$$

With $\mu = e^{rT} - 1 - \frac{e^{rT}}{2}V - \frac{e^{rT}}{6}W - \frac{e^{rT}}{24}X$, the risk-neutral moments of $r_T = \log(S_T/S_0)$ are

$$\text{Var}^{\mathbb{Q}}[r_T] = e^{rT}V - \mu^2, \quad (11)$$

$$\text{Skew}^{\mathbb{Q}}[r_T] = \frac{e^{rT}W - 3\mu e^{rT}V + 2\mu^3}{(e^{rT}V - \mu^2)^{3/2}}, \quad (12)$$

$$\text{Kurt}^{\mathbb{Q}}[r_T] = \frac{e^{rT}X - 4\mu e^{rT}W + 6\mu^2 e^{rT}V - 3\mu^4}{(e^{rT}V - \mu^2)^2}. \quad (13)$$

The last step assembles central, standardized moments from the raw contracts V, W, X : μ is the mean correction, the $-\mu^2$ and $-3\mu e^{rT}V + 2\mu^3$ terms recenter the second and third moments, and the denominators standardize. The signs in the put integrals are the odd/even parity of (7) made explicit: symmetric for V and X , antisymmetric for W .

Remark 3.3 (The $n = 2$ column, worked). The mechanism is one calculation repeated three times; here is $n = 2$ in full. With $H(S) = r^2 = \log^2(S/S_0)$,

$$H'(S) = \frac{2 \log(S/S_0)}{S}, \quad H''(S) = \frac{2(1 - \log(S/S_0))}{S^2}.$$

Substituting H'' into the replication identity (6) and evaluating at each strike K gives the weight $2(1 - \log(K/S_0))/K^2$ on the call side and, after $\log(K/S_0) = -\log(S_0/K)$ for $K < S_0$, the weight $2(1 + \log(S_0/K))/K^2$ on the put side (exactly the two integrands of V). Taking the $\mathbb{Q}$ -expectation turns each option payoff into its price, so $\mathbb{E}^{\mathbb{Q}}[r^2] = e^{-rT}V$ up to the cash and forward terms collected in μ . The columns for $n = 3$ and $n = 4$ are the same three steps with H'' from (7); only the number of terms grows.

Remark 3.4 (What “model-free” buys, and what it does not). No dynamics are assumed: not Heston, not SABR, not a jump specification. The moments are pinned by the surface alone, whatever process generated it, which is what makes $\text{Skew}^{\mathbb{Q}}$ comparable across names and across time. But “model-free” is a statement about the *continuum* of strikes in (7). A real surface is discrete and truncated, and the moment then depends on how the gaps are interpolated and the wings extrapolated. The model dependence removed from the dynamics reappears in the surface construction, which is why the extraction is only as trustworthy as the fitted surface beneath it.

4 The realized leg is noisy

The premium is a difference of two legs, and the realized leg is estimated from a finite return sample. That estimate is far noisier than the corresponding variance estimate, and ignoring the noise is the most common way to turn a real premium into a phantom one.

4.1 The standard error of sample skewness

For n log returns $r_1, \dots, r_n$ the sample skewness is

$$\widehat{\text{Skew}} = \frac{\frac{1}{n} \sum_i (r_i - \bar{r})^3}{\left(\frac{1}{n} \sum_i (r_i - \bar{r})^2\right)^{3/2}}. \quad (14)$$

Under i.i.d. normal returns the estimator is asymptotically unbiased and $\sqrt{n} \widehat{\text{Skew}} \xrightarrow{d} \mathcal{N}(0, 6)$, so the standard error is $\sqrt{6/n}$. The variance factor 6 is worth deriving, because it is exactly where the noise comes from, and because the derivation shows the noise is even worse on real data.

Where the 6 comes from. The argument is the central limit theorem applied to the right quantity, in three steps.

Step 1: the estimator is really an average. Write the returns as standardized ($\mathbb{E}[r] = 0$, $\mathbb{E}[r^2] = 1$). The denominator of $\widehat{\text{Skew}}$, the sample variance, tends to 1, so to leading order $\widehat{\text{Skew}}$ is just its numerator, $\frac{1}{n} \sum_i (r_i - \bar{r})^3$. Expanding the cube and keeping only the terms that survive ($\bar{r}$ is of order $1/\sqrt{n}$, so anything in $\bar{r}^2$ is negligible —

$$\frac{1}{n} \sum_i (r_i - \bar{r})^3 \approx \frac{1}{n} \sum_i r_i^3 - 3\bar{r} = \frac{1}{n} \sum_i (r_i^3 - 3r_i),$$

using $\bar{r} = \frac{1}{n} \sum_i r_i$ in the last step. So the sample skewness is, to leading order, the average of $r_i^3 - 3r_i$.

Step 2: the $-3r$ is the cost of estimating the mean. That correction did not come from nowhere: it is the $-3r_i^2 \bar{r}$ term of the expanded cube, and it exists only because the data are centered by the *sample* mean $\bar{r}$ rather than the true mean. Had the mean been known to be zero, the summand would be r^3 alone.

Step 3: the CLT gives the variance. An average of n i.i.d. terms, scaled by $\sqrt{n}$, is asymptotically normal with variance equal to the variance of one term. Here the term is $r^3 - 3r$, and for a standard normal (even moments $\mathbb{E}[r^2] = 1$, $\mathbb{E}[r^4] = 3$, $\mathbb{E}[r^6] = 15$)

$$\text{Var}(r^3 - 3r) = \mathbb{E}[(r^3 - 3r)^2] = \mathbb{E}[r^6] - 6\mathbb{E}[r^4] + 9\mathbb{E}[r^2] = 15 - 18 + 9 = 6.$$

The centering is what makes this small: the naive $\text{Var}(r^3) = \mathbb{E}[r^6] = 15$ is more than twice as large, and the $-6\mathbb{E}[r^4]$ cross term (the correlation between r^3 and r) is the cancellation that brings 15 down to 6.

This is a floor, not the typical case. The 6 rests on $\mathbb{E}[r^6] = 15$, which is specific to the normal; real returns are fat-tailed, their sixth moment is far larger, and the standard error is correspondingly worse than $\sqrt{6/n}$. The same argument one rung up, with the summand $r^4 - 6r^2 + 3$, gives $\text{Var} = 24$ and the $\sqrt{24/n}$ used for kurtosis later. and this is not a small number. At $n = 60$, one quarter of daily data, it is 0.32; at $n = 252$, a full year, still 0.15. A single quarter's realized skew of -0.5 thus carries a 95% interval of roughly $[-1.1, +0.1]$, and that is under the most favourable assumption possible. Real returns are fat-tailed, which inflates $\mathbb{E}[r^6]$ and pushes the true variance above 6, so $\sqrt{6/n}$ is a floor. The realized leg of a single-window comparison is often indistinguishable from zero.

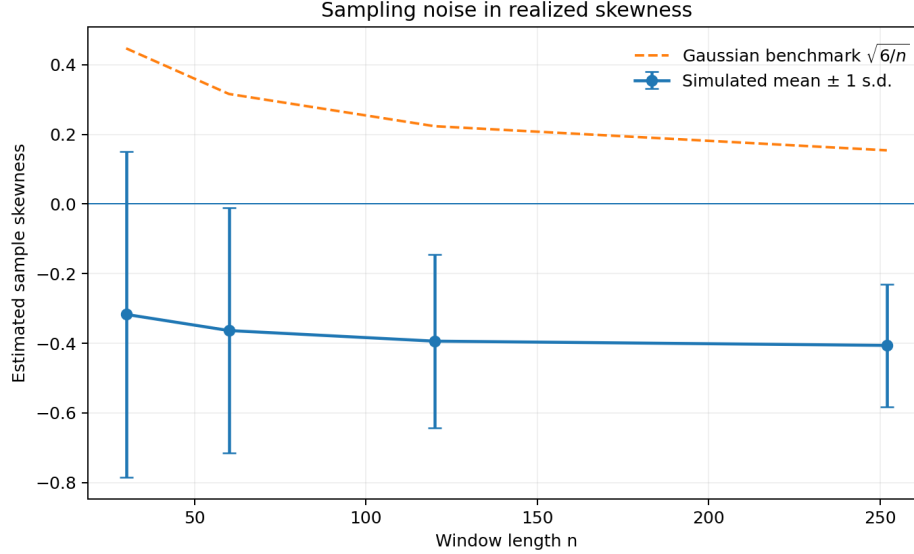


Figure 5: **Sampling noise in realized skewness.** Realized skewness is estimated from a finite sample and can therefore be extremely noisy, particularly over short windows. Under the Gaussian benchmark, the standard error of sample skewness is approximately $SE(\hat{\gamma}_1) \approx \sqrt{6/n}$. The dashed line shows this benchmark. The simulated estimates illustrate that increasing the observation window reduces sampling variation, while fat tails and dependence can make the effective uncertainty larger than the Gaussian benchmark.

The practical rule: never set a realized-skew number against implied skew from one window and call the difference a premium. The sampling error on the realized leg is comparable to the premium one is trying to measure. Longer windows, robust estimators, and cross-sectional ranking (below) are not refinements but requirements.

4.2 Robust alternatives

Because the third moment is dominated by the cubed tail observations, sample skewness inherits their instability, and the $\sqrt{6/n}$ standard error is only its best case: that figure assumed normal returns, and on fat-tailed data the sixth moment it rests on is far larger, so the true noise is worse. A Monte-Carlo check makes the gap concrete. At one year of daily data, the moment estimator has sampling standard deviation 0.15 on normal returns but 1.2 on Student- t returns with four degrees of freedom: the same estimator, an order of magnitude noisier, purely because the tail it cubes is heavier. On real return data the moment skew is barely an estimator at all.

This is why robust alternatives exist, and each is a different point on one trade-off: how much weight to give the tail that carries the signal against how much noise that tail injects. The moment estimator sits at one extreme, all signal and all noise. The others move along the axis by replacing cubed observations with quantiles, which are bounded and therefore cannot be blown up by a single extreme return.

Bowley skewness, $(Q_3 + Q_1 - 2Q_2)/(Q_3 - Q_1)$, measures asymmetry from the three quartiles alone: it asks whether the median sits closer to the upper or the lower quartile. It is bounded in $[-1, 1]$, and its noise is immune to the tails – in the same experiment its standard deviation is 0.085 on normal returns and 0.085 on the fat-tailed ones, unchanged, where the moment estimator moved by a factor of eight. The price is severe, and it is the point of tension in this section: by reading only

the quartiles, Bowley is nearly blind to the very tail asymmetry that skew is meant to capture. On a sample with true skew near -0.7 , driven by a rare large downside, Bowley reports about -0.05 . For a measure of tail asymmetry, ignoring the tail is not a small compromise.

Octile skewness, the same construction from the $1/8$ and $7/8$ quantiles, slides back along the axis toward the tail: it reads points further out, recovering some tail sensitivity while staying bounded and robust. It is the natural middle setting when the quartile version is too central and the moment version too noisy.

The realized semivariance ratio,

$$\frac{RS^- - RS^+}{RS^- + RS^+}, \quad RS^\mp = \sum_{i: r_i \leq 0} r_i^2, \quad (15)$$

is a different object, and worth including precisely because it is not a third moment. It splits realized variance into its down-move and up-move halves and asks which is larger – signed-variance asymmetry rather than distributional skewness. It is far more stable than the moment estimator because it squares rather than cubes, and it connects to the semivariance literature and to the downside-variance risk premium. But its being a different quantity is not a technicality: on the same left-skewed sample it can carry the opposite sign to the moment skew, because a distribution can be left-tail-skewed while realizing more total down-variance than up-variance, or the reverse. It answers a neighbouring question, not the same one, and should be read as a complement rather than a substitute.

Which to use follows from the trade-off, not from a ranking. For a single-name, single-window read where noise dominates, the bounded estimators keep the number meaningful. When the signal genuinely lives in the tail – crash-risk asymmetry, the case this note is about – the moment estimator is the only one that sees it, and the discipline is to lean on cross-sectional ranking and longer windows to control its noise rather than to discard the tail information a robust estimator throws away. The robust estimators buy statistical stability; they can also buy blindness to exactly the thing being measured, and knowing which is which is the whole skill.

4.3 Window length

Window choice is the same bias–variance trade-off in another guise. A short window (30 days) responds to regime shifts but is dominated by sampling noise; a long window (250 days) is stable but lags a genuine change in the tail. The defensible practice is not to pick one but to run both and read them as distinct signals: the short window carries the current regime, the long window anchors the structural level, and their disagreement is itself informative.

Realized skew has standard error $\approx \sqrt{6/n}$ under the best-case Gaussian assumption, comparable in size to the premium itself over a single window. The realized leg must be stabilized (longer windows, robust estimators (Bowley, semivariance ratio), or cross-sectional rankings that need only ordinal information) before any comparison to implied skew is meaningful.

5 What sustains the premium

The premium is not an accident of measurement; it is held open by economic forces that do not net out. Three of them account for most of the equity-index skew premium, and separating them matters because they tell you what you are actually short when you sell skew.

5.1 Flow: structural demand for the put wing

Pension funds, insurers, and structured-product issuers buy index puts systematically, to hedge liabilities and to fund note payoffs. That demand is concentrated at low strikes and it is persistent, so it steepens the put skew and pushes Skew^Q more negative than the realized asymmetry alone would warrant. This is a flow premium: it exists because the marginal buyer of the left wing is a hedger who is price-insensitive relative to the seller, and it persists because the hedging need is structural rather than opportunistic.

5.2 Jumps, and why realized skew under-counts them

The realized leg is measured from a finite sample of returns, and a finite sample systematically under-counts the tail asymmetry that the surface prices. Two mechanisms compound. First, sample skewness is bounded by sample size and starved of the rare large moves that dominate the third moment, so a quarter that happened not to contain a crash reports a near-symmetric skew even if the true distribution is badly left-skewed. Second, and more subtly, aggregation Gaussianizes: by the central limit theorem the distribution of a multi-day return is pulled toward the normal even when daily returns are skewed, because summing many increments washes out odd moments faster than even ones. Realized skew measured at one frequency is therefore not the realized skew relevant at another, and the horizon at which you measure changes the number. The surface, by contrast, prices the whole terminal distribution including the jumps the sample may not have drawn (so it “sees” tail risk that history has not yet shown, and the gap between the two is a genuine jump premium, not merely an estimation artifact).

5.3 Correlation: the index-only term

For an index there is a third force with no single-name analogue. Index skew is partly a correlation statement: in a sell-off, constituent correlations rise, so the index distribution has a fatter left tail than any constituent’s distribution does. Single-name realized skews do not aggregate up to index realized skew, because the aggregation itself is nonlinear in the correlation that stress inflates. This makes index risk-neutral skew structurally more negative than a correlation-blind aggregate of single-name skews (a term that is present for the index and absent for the single name).

The skew premium decomposes into a flow term (structural put demand), a jump term (the surface prices tail asymmetry the finite return sample under-counts, sharpened by the CLT washing skew out under aggregation), and, for indices only, a correlation term (stress correlation fattens the index left tail). Selling index skew is short all three at once (crash insurance, jump risk, and correlation — concentrated in the same regime. Selling single-name skew drops the correlation term and is mostly short idiosyncratic jump risk, a cleaner and smaller exposure; for some names (takeover candidates, retail-driven names) the flow runs the other way and the skew can even invert.

6 Trading the premium

6.1 The canonical vehicle: the risk reversal

The premium is expressed by trading one wing against the other. When the put wing is rich ($\mathcal{S}$ high, puts expensive in vol relative to calls) selling skew means *selling the rich put and buying the*

cheaper call at matched delta, the 25Δ put and 25Δ call being the standard choice. We describe the position by that action rather than by the sign of a quoted risk reversal, to keep the direction unambiguous.

6.2 The Greek profile, done carefully

Take the sell-skew position: short one 25Δ put, long one 25Δ call. The Greeks follow from the two legs directly.

Greek	Net sign	Source
Delta	$\approx +0.50$ (long)	short put (+0.25) + long call (+0.25)
Vega	≈ 0	long-call vega and short-put vega roughly cancel
Vanna	large, dominant	long the call wing, short the put wing
Gamma / tail	short the crash	the short put dominates if spot falls to its strike

Two points correct the arithmetic that is easy to get wrong. A short put has *positive* delta (the put's delta is negative; shorting flips the sign), so combined with the long call the position is *long* delta near +0.5, hedged by selling spot (not delta-neutral by construction. And the two vegas nearly cancel by matched delta, so this is not primarily a vol-level trade; what remains after delta and vega are handled is vanna.

6.3 What is really being traded

Strip a short risk reversal of its obvious exposures and see what is left. It carries delta, which is hedged away with the underlying, and vega, which very nearly cancels between the two legs on its own (a short OTM put and a long OTM call at symmetric deltas have similar vega, so selling one and buying the other leaves little, and what remains is neutralized explicitly. What does *not* cancel is vanna. Vanna, $\partial^2 V / \partial S \partial \sigma$, is the rate at which vega changes as spot moves, and it has *opposite* sign on the two wings: an OTM call gains vega as spot rises, an OTM put gains vega as spot falls. Being short the put and long the call, the two vannas add rather than offset. So the risk reversal is built, without trying to be, to isolate vanna: once delta and vega are handled, vanna is the dominant exposure that survives.

That this is the right exposure to hold is no accident, because vanna is the mechanical footprint of skew itself. By the equality of cross-derivatives, vanna is also $\partial \Delta / \partial \sigma$: how a position's delta shifts when volatility moves. A book with vanna is one whose delta hedge drifts precisely when spot and vol move together (which is what a skewed surface is, a surface on which spot and vol are correlated.

The position profits when the surface flattens, the put wing cheapening toward the call wing, and loses when it steepens, which is exactly what a sell-off does. The residual gamma is the sting: unlike vega, the two legs' gamma does not cancel, so a short risk reversal is short the crash, and its losses concentrate in the very regime that steepens skew. Collecting the skew premium is being paid to hold vanna and to be short the tail together.

6.4 Other vehicles, and an honest limit

Beyond the risk reversal, put and call vertical spreads express a wing view with bounded risk and better capital efficiency, at the cost of a messier Greek profile; and cross-asset relative value (selling a rich index's risk reversal against buying a cheap one) often mean-reverts more reliably than any single absolute level. One vehicle deserves a caveat, because the parallel to variance swaps is

tempting and false. A *skew swap* that pays on realized skewness does exist OTC, but there is no clean, model-free static replication of the third moment analogous to the $1/K^2$ log-contract strip that replicates variance. Realized third moment does not admit a payoff decomposition into a fixed strip of options held to expiry; skew swaps are therefore model-dependent, thinly traded, and far from the “path-independent, cleanly hedged” object a variance swap is. The clean replication result is special to the second moment, and it does not survive the step to the third.

6.5 Four situations of trading skew

Recall the convention from §1: a *risk reversal* is the call wing against the put wing, and selling it means selling the rich wing and buying the cheap one. With $S > 0$ the equity regime, puts richer than calls, “selling the risk reversal” is selling the put wing; where the skew is inverted the sides swap. Throughout, the position is described by which wing is sold rather than by the sign of a quoted number.

It is tempting to read the premium mechanically (implied skew rich against realized, therefore sell; flat against realized, therefore buy) as though the gap were a spring that must snap back. It is not. Each of the situations below is a *context*, and the same numerical signal means different things in different ones. A rich implied skew can be a mispricing that reverts, or the market correctly pricing a regime in which the tail is genuinely fatter; the screen cannot tell them apart, and treating the signal as a reversion rule is how one sells insurance into the event it was warning about. What follows names four contexts, each with its own reading and its own risk, precisely because the number alone does not carry the trade.

Rich implied, average realized: Implied skew in its top decile against a merely average realized skew: the market is overpaying for crash insurance relative to what has recently realized. Sell the risk reversal, delta-hedged and vega-neutral. Size for the tail crash that would validate the implied skew, because those losses fall in the same regime as the rest of a portfolio’s drawdowns.

Flat implied, negative realized: Implied skew in its bottom decile against a strongly negative realized skew: the market is underpricing a left tail that has been materializing. Buy the risk reversal, or the puts outright. Size to survive the theta bleed while waiting for the event that the skew is failing to price.

Single-name reverse skew: A name showing call skew during takeover speculation is a trap. The positive skew may be correctly pricing takeover optionality, so selling it is selling the right tail: modest premium collected against a large loss on deal news. The screen flags it as rich; the structure is a short position in exactly the event most likely to occur.

Cross-sectional: Rank the constituents by implied-minus-realized skew, sell the rich, buy the cheap, delta- and beta-neutral. This rests on the ordinal signal, which is more stable than any absolute level, and it is the most defensible of the four. Its risk is crowding: when many desks run the same screen, the unwinds correlate.

Situation	Signal	Trade	Principal risk
Rich implied	Top-decile $\mathcal{S}$, average realized	Sell the risk reversal	Tail crash validates the skew; losses coincide with portfolio drawdowns
Flat implied	Bottom-decile $\mathcal{S}$, strongly negative realized	Buy the RR, or puts outright	Theta bleed while waiting for the event
Reverse skew	Call skew in a takeover name	(usually pass)	Selling the right tail: small premium, large loss on deal news
Cross-sectional	Rank implied—realized skew	Sell rich, buy cheap, delta/beta neutral	Crowding: correlated unwinds

The canonical skew trade is a delta-hedged, vega-neutral short risk reversal, which is fundamentally a vanna trade: long vanna, short the crash, paid to hold both. Unlike variance, the third moment has no clean model-free replication, so a skew swap is a model-dependent OTC object, not the cleanly hedged instrument its name suggests. Cross-sectional and relative-value expressions rest on more stable signals than the absolute level.

7 Questions the framework should answer

A framework is only as good as the questions it survives. Five recur.

- ***If selling skew has positive expected return, why does the premium persist?*** Because it is compensation for a left-tail loss correlated with broad market drawdowns (a negative-skew P&L of small wins and rare large losses, with poor Sharpe exactly in stress and no diversification benefit when the portfolio most needs one. The supply of sellers is limited by that risk, and the demand (structural hedging flow) is large and persistent, so the premium clears above fair value and stays there. Many desks do sell skew; they size for survival rather than Sharpe, and the premium survives because the flow does.
- ***How does one tell a tradeable mispricing from a genuine regime shift in tail risk?*** By whether the implied skew is moving *alone* or *in company*. In a mispricing the implied skew dislocates and nothing corroborates it: realized skew does not follow across multiple windows, cross-sectional peers do not move, and the flow and macro indicators (credit spreads, put/call flow, term-structure inversion) stay put. It is a one-market dislocation, which is why it tends to revert. In a regime shift the same implied move is confirmed everywhere at once: realized skew steepens, peers move together, credit widens, flows turn. Nothing is dislocated, because the whole complex has repriced the tail together, and there is no reversion to trade.

The distinction requires watching many indicators, not because the trade needs many signals but because it needs the *context* the others provide. A desk looking at a single number (implied skew rich against one realized estimate) cannot make the call at all: with nothing to compare against, an isolated dislocation and a corroborated repricing look identical. The danger is not that the signal stands alone, that is the setup one wants, but that the observer does, and mistakes a regime shift for the mispricing it resembles.

- **How does one build a reliable signal from an estimator with standard error $\sqrt{6/n}$?** By not relying on the raw estimator: longer windows for the baseline, robust estimators (Bowley, semivariance ratio) for confirmation, and above all cross-sectional ranking, which needs only ordinal information and is far less sensitive to absolute estimation noise. The best signals combine all three and trade the ranking, not the level.
- **Why is index skew structurally steeper than single-name skew?** The three forces of §5: structural put-hedging flow concentrated in index products; rising correlation in stress, which fattens the index left tail beyond any constituent's; and idiosyncratic single-name upside (takeovers, earnings beats) that creates call demand and flattens or inverts single-name skew. Index skew is correlation skew plus a correlation-blind aggregate of single-name skews plus flow.

Given $\text{Skew}^{\mathbb{Q}} = -2.1$ from BKM and a realized 90-day skew of -0.5 , what do you do?

First, distrust the inputs in the specific ways this note identifies: check that the $\mathbb{Q}$ -skew was computed on a wide enough strike grid, since truncation biases it *toward* zero and a -2.1 off a narrow grid understates the true asymmetry; confirm the realized leg with a robust estimator and a second window, since a single -0.5 has a standard error near 0.3; and read the flow and regime context. Only if all of these agree is the gap real. Then trade it small, a short risk reversal sized to survive a three-sigma left tail, with an explicit OTM put to cap the loss, at a fraction of normal size, because rich-skew regimes are disproportionately the ones that precede the crash that validates the skew.

Part II

Remarks about trading other moments

8 Why variance, not volatility

Before pushing the moment programme upward to skew and kurtosis, it is worth being explicit about a choice made silently at the second moment, because the same nonlinearity that motivates it is what will break the higher rungs. We speak of the volatility premium (investors fear volatility, and that is the economically meaningful quantity) but every clean instrument is built in *variance*. The reason is that volatility is the square root of the second moment, and that square root is the first place the machinery stops being clean.

8.1 Variance is additive; volatility is not

Variance over $[0, T]$ is the sum of the variances of its subintervals: for independent increments, $\sigma^2 T = \sigma_1^2 t_1 + \sigma_2^2 t_2 + \dots$. Volatility carries no such decomposition ($\sigma\sqrt{T}$ does not split into a sum). Any operation that combines, decomposes, or rescales in time is therefore natural in variance and awkward in volatility. This is not cosmetic: the entire event-volatility decomposition, in which an option's total implied variance splits into a diffusive piece and an event piece, exists *only* because variance is additive. In volatility units the decomposition cannot even be written.

8.2 Variance replicates cleanly; volatility does not

The deeper reason is replication. Realized variance admits a model-free static replication (the $1/K^2$ log-contract strip) so a variance swap can be priced and hedged off the option surface with no

model of the dynamics. Realized volatility, its square root, admits no such thing: there is no static portfolio of options whose payoff is realized volatility, because the square root is nonlinear and the replication argument that works for variance does not survive it. A volatility swap consequently requires a convexity correction that depends on the vol-of-vol (that is, a model) where the variance swap requires none. Variance is the quantity the surface can replicate; volatility is one nonlinear step removed from it.

8.3 The square root carries a Jensen wedge

The two are not even related by a simple square root at the level of fair values. Because $\sigma = \sqrt{\sigma^2}$ is concave, Jensen's inequality gives $\mathbb{E}[\sqrt{\sigma^2}] \leq \sqrt{\mathbb{E}[\sigma^2]}$, so the fair volatility-swap strike is strictly below the square root of the fair variance-swap strike, with the gap governed by the vol-of-vol. The convexity that makes volatility awkward to replicate is the same convexity that separates the two fair values, and both trace to the concavity of the square root.

8.4 The VIX methodology made exactly this switch

The index history is the market having reached the same conclusion. The original VIX (1993, now ticker VXO) was a *volatility* construction: it averaged the Black–Scholes implied volatilities of a handful of near-the-money options. Two problems followed directly from the choice. It was model-dependent (inverting prices into implied vols requires Black–Scholes) and, using only ATM options, it ignored the entire skew. In 2003 the CBOE replaced it with the current VIX, a *variance* construction: the $1/K^2$ -weighted strip across all strikes, square-rooted only at the very end to quote the result in volatility units. The new formula is model-free, being a direct portfolio of option prices rather than an average of model-inverted vols, and it uses the whole surface, capturing the skew the old version discarded. The switch was made for precisely the reasons above (additivity, clean replication, no model dependence) and the square-root-at-the-end quoting is the tell: the object is built in variance and only dressed as volatility for display.

One trades the volatility premium but builds the instrument in variance, because variance adds over time, replicates model-free off the $1/K^2$ strip, and carries no Jensen correction, while volatility (its square root) fails all three. The 1993 → 2003 VIX change is the market making this choice explicitly. And the lesson previews the next section: the square root is the first nonlinearity to break the clean machinery at the second moment. The higher moments break it further, and §9 is where that breakage becomes a ceiling.

9 Where the moment ladder breaks

The programme so far is a ladder. The second moment gives the variance premium; the third gives the skew premium; each is a $\mathbb{Q}$ -object from BKM against a $\mathbb{P}$ -object from returns, and each is independently tradeable. Nothing in the construction stops at three (BKM gives Kurt^Q from the same surface, and one could define a kurtosis premium identically. The question this note builds toward is why one should not: why the ladder stops paying for itself above the third rung. Two forces degrade every rung, and both accelerate.

9.1 Measurement noise doubles with each rung

The realized leg of the m -th moment is estimated from m -th powers of returns, and higher powers are dominated ever more heavily by the rare large observations, so the estimator variance grows fast. Concretely, against the $\sqrt{6/n}$ standard error of sample skewness, sample excess kurtosis carries $\sqrt{24/n}$ (Monte-Carlo confirmed at 0.30 for $n = 252$ against $\sqrt{24/252} = 0.31$. The ratio is exactly $\sqrt{24/6} = 2$: each step up the ladder roughly *doubles* the sampling noise of the realized leg. By the fourth moment the realized estimate over any tradeable window is essentially uninformative on its own, and the fifth is worse still.

9.2 Truncation bias worsens, and then inverts the sign

The extraction leg degrades in parallel, and more alarmingly. The BKM kernels carry rising powers of $\log(K/S_0)$ (linear for variance, quadratic for skew, cubic for kurtosis) so each higher moment loads the far wing more heavily and is more sensitive to the truncation every real strike grid imposes. The ground-truth test of §3 makes the progression concrete. On the same crash-prone density, truncating to a realistic listed range leaves skew at 45% then 23% of its true value (damaged but still the right sign). Kurtosis does worse than shrink:

Strike range used	extracted Skew ^Q	extracted excess Kurt ^Q
True	−1.41	+3.84
Near-full support	−1.40	+3.84
± 30% around the money	−0.64	−0.08
± 20% around the money	−0.33	−1.20

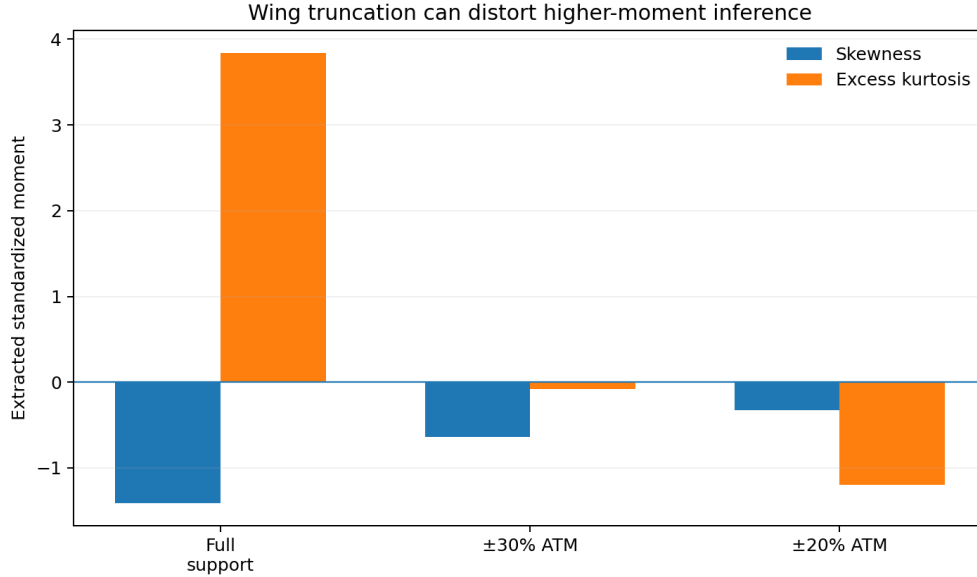


Figure 6: **Wing truncation can distort higher-moment inference.** The same underlying risk-neutral distribution is considered using progressively narrower regions of the option surface. Restricting the surface to $\pm 30\%$ and then $\pm 20\%$ around the money removes increasingly important tail information. The estimated skewness is substantially attenuated, while the estimated excess kurtosis eventually changes sign. This illustrates why higher moments are not only noisier, but also increasingly sensitive to the portion of the distribution that cannot be observed reliably through a finite option surface.

Truncation drives the extracted excess kurtosis through zero and *negative*: a strongly fat-tailed distribution is reported as thinner-than-normal. The sign is wrong, not merely the magnitude. A kurtosis premium computed off a standard listed grid can point the opposite way to the truth, which for a tradeable signal is not noise but poison.

9.3 The premium stops being distinct

Even granting perfect measurement, the higher moments are not economically independent of the lower ones. On equity surfaces, kurtosis richness co-moves strongly with skew and variance richness (the same crash-fear flow that steepens the skew fattens the wings (so a putative kurtosis premium is largely a repackaging of the skew and variance premia already being traded. The marginal information in the fourth moment, over and above the second and third, is small, and it shrinks further with each rung. The ladder does not merely get noisier; its rungs get less distinct.

The moment ladder is not infinite. Measurement noise on the realized leg roughly doubles with each rung ($\sqrt{6/n}$ for skew, $\sqrt{24/n}$ for kurtosis); truncation bias on the extracted leg worsens and, for kurtosis, inverts the sign on a realistic strike grid; and the higher moments co-move with the lower ones, so the marginal premium shrinks. Variance (rung two) is robust and the variance premium is the workhorse; skew (rung three) is tradeable with the care this note describes; kurtosis (rung four) is at best a weak confirming signal and easily a misleading one; beyond it is noise. Knowing where the ladder stops is part of using it.

10 Synthesis

The skew risk premium is the variance premium’s construction carried one moment up. Implied skew (the surface slope, a $\mathbb{Q}$ -measure summary of risk-neutral asymmetry, read model-free from option prices through Bakshi–Kapadia–Madan) is more extreme than realized skew, the $\mathbb{P}$ -measure third moment of returns. The gap is a genuine premium, held open by structural put-hedging flow, by the jump asymmetry a finite return sample under-counts and aggregation Gaussianizes away, and, for indices, by the correlation that fattens the left tail in stress. It is traded through a delta-hedged, vega-neutral short risk reversal, which is at bottom a vanna trade: long vanna, short the crash, paid to hold both. Its realized leg is noisy enough that a single-window estimate is barely distinguishable from zero, so the honest signal leans on robust estimators and cross-sectional ranking rather than absolute levels.

And the ladder that produced it stops. Each rung doubles the measurement noise and worsens the truncation bias (to the point that a naively extracted kurtosis can carry the wrong sign) while the marginal premium shrinks as higher moments collapse into the lower ones already traded. Variance is the robust rung and skew is the tradeable one; past the third moment the construction still runs but no longer pays. The discipline the note argues for is the same one that runs through the variance work: know which quantity your surface can actually measure, which premium is genuinely distinct, and where the clean machinery quietly stops being clean.

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